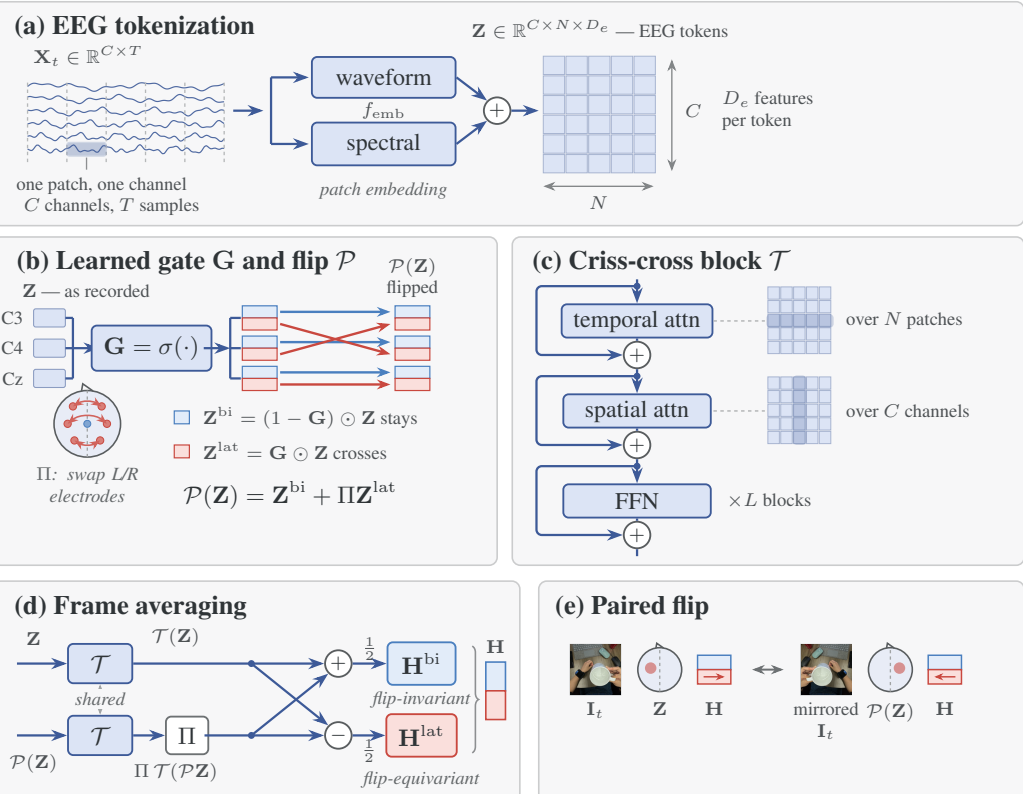

(a) EEG tokenization
$\mathbf{X}_t \in \mathbb{R}^{C\times T}$
$\mathbf{Z} \in \mathbb{R}^{C\times N\times D_e}$ — EEG tokens
waveform
f_{emb}
spectral
patch embedding
C
D_e features per token
N
one patch, one channel
C channels, T samples
(b) Learned gate $\mathbf{G}$ and flip $\mathcal{P}$
$\mathbf{Z}$ — as recorded
$\mathcal{P}(\mathbf{Z})$ flipped
C3
C4
Cz
$\mathbf{G} = \sigma(\cdot)$
$\mathbf{Z}^{\text{bi}} = (1-\mathbf{G})\odot\mathbf{Z}$ stays
$\mathbf{Z}^{\text{lat}} = \mathbf{G}\odot\mathbf{Z}$ crosses
Π: swap L/R electrodes
$\mathcal{P}(\mathbf{Z}) = \mathbf{Z}^{\text{bi}} + \Pi\mathbf{Z}^{\text{lat}}$
(c) Criss-cross block $\mathcal{T}$
temporal attn
over N patches
spatial attn
over C channels
FFN
$\times L$ blocks
(d) Frame averaging
$\mathbf{Z}$
$\mathcal{T}(\mathbf{Z})$
shared
$\mathcal{P}(\mathbf{Z})$
$\Pi\,\mathcal{T}(\mathcal{P}\mathbf{Z})$
$\frac{1}{2}$
$\mathbf{H}^{\text{bi}}$
flip-invariant
$\mathbf{H}^{\text{lat}}$
flip-equivariant
$\mathbf{H}$
(e) Paired flip
$\mathbf{I}_t$
$\mathbf{Z}$
$\mathbf{H}$
mirrored $\mathbf{I}_t$
$\mathcal{P}(\mathbf{Z})$
$\mathbf{H}$